

Chibuike Robinson Umeike

PhD Candidate — AI/ML Researcher — Computer Vision & Foundation Models

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PROFESSIONAL SUMMARY

AI/ML researcher with expertise in multimodal foundation models, computer vision, and scalable production systems. Proven track record at Argonne National Laboratory developing vision-language models achieving **135% performance improvement** and **96.6% reduction in hedging behavior**. Designed real-time CV systems processing **333 images per second**. Published at top-tier venues (NeurIPS, IEEE ISBI). Seeking industry role leveraging computer vision, deep learning, HPC infrastructure, and end-to-end ML pipeline development.

TECHNICAL SKILLS

- **Languages:** Python, C#, C++, Java, MATLAB, Bash
- **Development Tools:** Git, Plotly, NumPy, Pandas, LaTeX
- **ML/AI Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, vLLM, LoRA, Scikit-Learn, OpenCV, VERL
- **Production ML:** HPC systems, Real-time inference, Data curation pipelines, Multi-GPU training
- **Computer Vision:** YOLO, RT-DETR, SSD, ResNet, CLIP, Object detection, Segmentation, Image classification, etc.
- **Foundation Models:** LLaVA, Qwen, Llama, Vicuna, Large Language Models, Vision-Language Models, Reinforcement Learning (GRPO/PPO)

PROFESSIONAL EXPERIENCE

- **Argonne National Laboratory (U.S. Department of Energy)** Lemont, IL
Research Technical Aide - AI/ML for Scientific Computing May 2024 – Aug 2025
 - **Foundation Models for Scientific Imaging (Summer 2025):** Engineered domain-aware vision-language models (Qwen-2.5VL, Llama-3.2) using reinforcement learning (GRPO) for X-ray ptychography analysis. Managed end-to-end data pipeline processing synchrotron data from Advanced Photon Source. Developed PtychoBench benchmark for evaluating foundation model adaptation. *Impact: First author, NeurIPS 2025 AI4MAT Workshop.*
 - **Vision-Language Models for Biomedical Applications (Summer 2024):** Built comprehensive data curation pipeline processing 42,673 scientific articles, generating 171K+ captions and 165K+ images. Fine-tuned LLaVA models (13B parameters) using LoRA on 50,882 image-text pairs. Achieved 135% performance improvement and 96.6% hallucination reduction over baselines. *Impact: Oral presentation IEEE ISBI 2025 (≈20% acceptance), invited to Trillion Parameter Consortium (90+ institutions).*
 - **Technical Leadership:** Presented research to DOE leadership and Argonne scientific staff. Collaborated with Computing, Environment and Life Sciences Directorate. Utilized ALCF HPC infrastructure for large-scale model training, and evaluation.
- **The University of Alabama** Tuscaloosa, AL
Doctoral Researcher - AI for Disaster Resilience & Smart Cities June 2021 – Present
 - **Real-Time Computer Vision System:** Developed end-to-end deep learning pipeline using YOLO, RT-DETR, and ResNet for automated building damage assessment. Deployed system processing 333 images/second, reducing evaluation time from weeks to minutes. *Impact: 2nd place NSBE National Convention (15,000+ attendees), ASCE Editor's Choice Award.*
 - **Multi-Institutional Research Leadership:** Led technical coordination for NIST Disaster and Failure Studies Program consortium (NIST, Johns Hopkins, Colorado State, South Alabama). Delivered bi-weekly progress updates to federal stakeholders. Project showcased at NIST AI Day 2024.
 - **IoT Smart Cities Platform:** First application of OIIE international standards to real-time urban flood monitoring. Deployed wireless sensor network processing 25,000+ data points. Built mobile app and real-time dashboard for emergency response. *Impact: 2 first-place innovation awards, TISDIC 2023 publication.*
- **VFD Tech Limited (VFD Group Plc)** Lagos, Nigeria
IT Project Manager June 2020 – June 2021
 - **Product Management:** Managed 25+ technology projects with cross-functional teams (10+ developers). Led second major release of V mobile banking app (1M+ Android downloads), introducing proximity payment and cardless withdrawal features.
 - **Platform Development:** Oversaw Banking-as-a-Service (BaaS) platform deployment serving 20+ corporate clients. Coordinated user acceptance testing and stakeholder communication.

EDUCATION

- **University of Alabama** Tuscaloosa, AL
PhD, Civil Engineering (AI/Computer Vision focus) — GPA: 3.85/4.0 Expected May 2026
 - Dissertation: *Automated Framework for Rapid Post-Disaster Infrastructure Assessment Using Deep Learning*
 - Advisor: Dr. Thang Dao (Associate Editor, ASCE Journal of Structural Engineering)
- **University of Alabama** Tuscaloosa, AL
MS, Computer Science — GPA: 3.85/4.0 Dec 2024
- **University of Alabama** Tuscaloosa, AL
MS, Civil Engineering — GPA: 4.0/4.0 Dec 2023

KEY PROJECTS & OPEN SOURCE

- **PtychoBench - Adapting Foundation Models for X-ray Imaging** (github.com/crumeike/PtychoBench): Novel benchmark for evaluating foundation model adaptation to X-ray ptychography. Domain-aware VLMs and LLMs (Llama-3.1, Llama-3.2-vision). Published NeurIPS 2025 AI4MAT.
- **LUCID Biomedical VLM** (github.com/crumeike/lucid-biomedical-vlm): Vision-language models for scientific imaging. Dataset curation (171K captions), LoRA fine-tuning, LLM-as-judge evaluation. Published IEEE ISBI 2025.
- **Smart Cities IoT Platform** (github.com/crumeike/smart-cities-iot): Real-time flood monitoring with OIIE standards. Edge computing, mobile alerts, decision support dashboard. Published TISDIC 2023.
- **Post-Disaster Assessment AI**: Real-time building damage detection using YOLO/RT-DETR. Processes 333 fps. Multi-institutional NIST consortium project. See GitHub: github.com/crumeike

SELECTED PUBLICATIONS (10 TOTAL, 7 FIRST-AUTHOR, 2 SOLE-AUTHOR)

- **R. Umeike**, N. Getty, X. Yin, Y. Jiang, "Adapting General-Purpose Foundation Models for X-ray Ptychography in Low-Data Regimes," *NeurIPS 2025 AI4MAT Workshop*, 2025.
- **R. Umeike**, N. Getty, F. Xia, R. Stevens, "Scaling Large Vision-Language Models for Enhanced Multimodal Comprehension in Biomedical Image Analysis," *IEEE ISBI 2025 (Oral)*, 2025.
- **R. Umeike**, "Contact-Based Grasp Control and Inverse Kinematics for a Five-Fingered Robotic Hand," *IEEE ICMRE 2025*, pp. 116-121, 2025.
- **R. Umeike**, "Comparative Crime Analysis and Prediction Using Machine Learning Algorithms: Assessing the Tools and Addressing the Threats," *IEEE ICTAI 2023*, pp. 135-142, 2023.
- **R. Umeike**, T. Dao, S. Crawford, "State-of-the-Art Review of Resilient Smart Cities: Progress and Challenges," *Urban Governance*, vol. 5, no. 2, pp. 222-239, 2025.
- W. Wang, J. van de Lindt, et al., **R. Umeike**, "Application of Multidisciplinary Community Resilience Modeling to Reduce Disaster Risk," *J. Performance of Constructed Facilities* (ASCE Editor's Choice), vol. 38, no. 3, 2024.
- *Full publication list*: scholar.google.com/citations?user=IeJsfl8AAAAJ

HONORS & RECOGNITION

- **NeurIPS AI4MAT Travel Grant** - Competitive award for conference presentation (2025)
- **2nd Place, NSBE National Technical Research Exhibition** - 15,000+ attendees (2025)
- **ASCE Editor's Choice Award** - Awarded to fewer than 2% of all 200,000+ ASCE publications (2024)
- **Argonne Sustainable Research Pathways Fellowship** - Competitive fellowship for placement in the DOE (2024)
- **Invited Speaker, Trillion Parameter Consortium** - Global AI consortium, 90+ institutions (2024)

ADDITIONAL IMPACT

- **Peer Review Service**: over 46 reviews for top-tier journals/conferences (IEEE ISBI, IJDRR (Ranked #1 in emergency management), Engineering Applications of AI, ASCE journals)
- **Media Features**: Argonne National Laboratory, ALCF, TPC Seminar Series, ASCE Civil Engineering Source
- **Invited Talks**: NIST AI Day 2024, Argonne Learning-off-the-Lawn, TPC Global Seminar (90+ institutions & 4 continents)